CYBERSRS

Wireless Network Access Control Requirements

Software Requirements Specification

DRAFT — human review required

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AI-generated advisory document. Review by a qualified professional is required.

Automated checks do not prove completeness or semantic correctness. SDLC suitability scores are rule-based points, not probabilities.

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1. Introduction

Original project description

Build a campus wireless network access-control requirements specification. Students and guests may access the internet but must not access the internal staff network. Network administrators may manually isolate a device and restore its access. Security operators may review and acknowledge alerts but may not change access policies. The system must record administrator policy changes and retain audit records for 30 days. The planning population is 2000 registered devices; concurrency is not yet decided. Do not automatically change firewall rules or isolate devices. Requirements evolve frequently and stakeholders provide weekly stakeholder feedback. Budget, delivery schedule, response-time thresholds and mandatory regulatory frameworks are undecided and must remain open decisions.

Summary: This specification records the project purpose, stakeholder constraints and unresolved decisions stated in the source evidence below.

Items

Stakeholder project context

Detail: Build a campus wireless network access-control requirements specification.

Status: confirmed

Evidence ids: E-001

Stakeholder source statement E-002

Detail: Students and guests may access the internet but must not access the internal staff network.

Status: confirmed

Evidence ids: E-002

Stakeholder source statement E-003

Detail: Network administrators may manually isolate a device and restore its access.

Status: confirmed

Evidence ids: E-003

Stakeholder source statement E-004

Detail: Security operators may review and acknowledge alerts but may not change access policies.

Status: confirmed

Evidence ids: E-004

Stakeholder source statement E-005

Detail: The system must record administrator policy changes and retain audit records for 30 days.

Status: confirmed

Evidence ids: E-005

Unresolved stakeholder decision

Detail: The planning population is 2000 registered devices; concurrency is not yet decided.

Status: unresolved

Evidence ids: E-006

Out of scope: stakeholder exclusion

Detail: Do not automatically change firewall rules or isolate devices.

Status: confirmed

Evidence ids: E-007

Stakeholder project context

Detail: Requirements evolve frequently and stakeholders provide weekly stakeholder feedback.

Status: confirmed

Evidence ids: E-008

Unresolved stakeholder decision

Detail: Budget, delivery schedule, response-time thresholds and mandatory regulatory frameworks are undecided and must remain open decisions.

Status: unresolved

Evidence ids: E-009

Open issues: What is the defined concurrency level for device access?; How will frequent requirement changes be tracked or managed in the development lifecycle?; Which specific regulatory frameworks must be considered if any are later identified?; What is the process for handling device isolation when a device is found to be malicious?; Will automated alerts trigger any actions, and if so, what is the threshold or condition?; The planning population is 2000 registered devices; concurrency is not yet decided.; Budget, delivery schedule, response-time thresholds and mandatory regulatory frameworks are undecided and must remain open decisions.

2. Functional requirements

Applicability reason: The evidence explicitly states that students and guests may access the internet but must not access the internal staff network, establishing a clear functional boundary for wireless access control. This directly supports a functional requirement in access control behavior. The source passage explicitly states that network administrators may manually isolate a device and restore its access, defining a functional administrative exception path within wireless network access control. The system shall not automatically change firewall rules or isolate devices, prohibiting automated actions and restricting to manual control only.

Requirements

FR-001 - Students and guests may access the internet but must not access the internal staff network

Statement: Students and guests may access the internet but must not access the internal staff network.

Rationale: This establishes a clear boundary between public-facing internet access and restricted internal staff network access, ensuring that unauthorized users do not gain entry to sensitive internal systems.

Priority: must

Dimensions: security; availability; operations

Basis: explicit

Evidence ids: E-002

Acceptance criteria

Given: A student or guest is authenticated and connected to the wireless network using a valid public identity.

When: The user attempts to access the internet via standard web browsing services.

Then: The system allows successful access to publicly available internet content.

Method: test

Observable evidence: Record of successful page load from a public website, such as google.com.

Given: A student or guest is authenticated and connected to the wireless network using a valid public identity.

When: The user attempts to access internal staff network resources via IP address or domain name (e.g., internal.staff.company.com).

Then: The system denies access and no product message is returned; traffic is blocked as evidenced by an external test capture of the network response state.

Method: inspection

Observable evidence: Export of a test-instrumented network response state capture showing denied access to internal staff network resources, captured at the time of request and verified against known firewall rule enforcement.

Unresolved decisions: Whether the system must enforce this boundary at the application layer or only via network-level filtering.; The specific mechanism for detecting and blocking attempts to reach internal staff network resources.

FR-002 - Network administrators may manually isolate a device and restore its access

Statement: Network administrators may manually isolate a device and restore its access.

Rationale: This action defines an explicit administrative exception path for device management, allowing manual intervention when necessary without requiring automated or policy-driven isolation mechanisms.

Priority: must

Dimensions: security; operations

Basis: explicit

Evidence ids: E-003

Acceptance criteria

Given: A network administrator logs into the system with valid administrative credentials.

When: The administrator selects a device to isolate via the manual isolation interface.

Then: The system isolates the device from wireless access and updates the access control state accordingly.

Method: test

Observable evidence: Capture the system response state after manual isolation via API call and validate it against expected access control update.

Given: A network administrator logs into the system with valid administrative credentials.

When: The administrator selects a device that has been isolated to restore access via the manual restoration interface.

Then: The system restores wireless access for the device and updates the access control state accordingly.

Method: test

Observable evidence: Capture the system response state after manual restoration of wireless access for the device, including the updated access control state and the timestamp of the action.

Unresolved decisions: The specific interface or UI component used for isolation and restoration is not defined; this remains an implementation parameter.

FR-003 - Do not automatically change firewall rules or isolate devices

Statement: The system shall not automatically change firewall rules or isolate devices.

Rationale: Explicitly prohibits automated actions such as isolation or rule modification, ensuring that all such operations are performed manually by authorized personnel.

Priority: must

Dimensions: security

Basis: explicit

Evidence ids: E-007

Acceptance criteria

Given: The system is configured to automatically modify firewall rules or isolate a device in response to an event

When: An automated action such as rule modification or isolation is initiated by the system

Then: The action shall be denied and no change to firewall configuration or device state shall occur

Method: test

Observable evidence: Capture of unchanged firewall configuration and unmodified device state before and after the attempted action

Related requirement ids: SEC-001

3. Non-functional requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-002; FR-003

4. Security, privacy and compliance

Applicability reason: Security operators may review and acknowledge alerts but may not change access policies, restricting operator permissions to monitoring only. The evidence explicitly states a requirement for recording administrator policy changes and retaining audit records for 30 days, which directly supports a security-related data retention and audit control.

Requirements

SEC-001 - Security operators shall be permitted to review and acknowledge alerts

Statement: Security operators may review and acknowledge alerts but may not change access policies.

Rationale: The source passage explicitly states that security operators may review and acknowledge alerts but may not modify access policies, which defines their authorized actions in the system.

Priority: must

Dimensions: security

Basis: explicit

Evidence ids: E-004

Acceptance criteria

Given: A security operator accesses an alert dashboard

When: The operator views a network anomaly report

Then: The operator can acknowledge the alert and receive confirmation of acknowledgment

Method: inspection

Observable evidence: Record of acknowledged alert with timestamp and operator identity

Given: An operator receives an alert with a visible status indicator.

When: The operator reviews the alert and acknowledges it via the system interface.

Then: The system updates the alert status to 'acknowledged' and retains the original policy unchanged; no policy modification occurs.

Method: test

Observable evidence: Capture of the system response showing alert status updated to ' acknowledged' with no change in access policies, as verified by a test-instrumented state capture of the alert object's properties.

Given: A security operator receives an alert with a policy violation.

When: The operator attempts to modify an access policy through the system interface.

Then: The system denies the modification request and returns a response indicating that policy modifications are not permitted for operators.

Method: test

Observable evidence: Capture of the system's direct response shows a denial of the modification action, with no change to the access policies in storage or configuration.

SEC-002 - The system shall record administrator policy changes and retain audit records for 30 days

Statement: The system shall record administrator policy changes and retain audit records for 30 days.

Rationale: Establishes a data retention requirement with an audit purpose to ensure accountability and traceability of administrative actions affecting network access policies.

Priority: must

Dimensions: security; auditability

Basis: explicit

Evidence ids: E-005

Acceptance criteria

Given: An administrator modifies a network access policy, such as changing authentication requirements or access permissions.

When: The system records the change and stores it in an audit log.

Then: The audit record is retained for 30 days and remains accessible for review by authorized personnel.

Method: inspection

Observable evidence: Inspect the audit log to confirm that a policy change event was recorded with timestamp, actor identity, action type, and before/after state.

Unresolved decisions: The specific format of the audit record (e.g., JSON, CSV) is not specified in the evidence.; The source gives a retention duration; minimum/maximum interpretation and deletion timing remain unconfirmed.

Related requirement ids: FR-001

5. Data requirements

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-002; FR-003; SEC-001; SEC-002

6. Network and interfaces

Applicability reason: Existing canonical requirements, where applicable, are linked below; this section adds no separate obligation. Unresolved stakeholder decisions remain open.

Related requirement ids: FR-001; FR-002; FR-003; SEC-001; SEC-002

7. Compliance and policy applicability matrix

Open decisions: What is the defined concurrency level for device access?; How will frequent requirement changes be tracked or managed in the development lifecycle?; Which specific regulatory frameworks must be considered if any are later identified?; What is the process for handling device isolation when a device is found to be malicious?; Will automated alerts trigger any actions, and if so, what is the threshold or condition?

8. Architecture

Summary: Proposed logical components and access-decision paths for source-stated boundaries; unresolved implementation choices remain open.

Items

Actor: Students and guests

Detail: Proposed logical endpoint for the source-stated actor: Students and guests.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-001

Component name: Actor: Students and guests

Responsibility: Represent the source-stated actor: Students and guests.

Resource: internet

Detail: Proposed logical endpoint for the source-stated resource: internet.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-001

Component name: Resource: internet

Responsibility: Represent the source-stated resource: internet.

Resource: internal staff network

Detail: Proposed logical endpoint for the source-stated resource: internal staff network.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-001

Component name: Resource: internal staff network

Responsibility: Represent the source-stated resource: internal staff network.

Actor: Network administrators

Detail: Proposed logical endpoint for the source-stated actor: Network administrators.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-002

Component name: Actor: Network administrators

Responsibility: Represent the source-stated actor: Network administrators.

Actor: Security operators

Detail: Proposed logical endpoint for the source-stated actor: Security operators.

Status: proposed

Evidence ids: E-004

Requirement ids: SEC-001

Component name: Actor: Security operators

Responsibility: Represent the source-stated actor: Security operators.

Actor: Administrator

Detail: Proposed logical endpoint for the source-stated actor: Administrator.

Status: proposed

Evidence ids: E-005

Requirement ids: SEC-002

Component name: Actor: Administrator

Responsibility: Represent the source-stated actor: Administrator.

Access Control Engine

Detail: Proposed logical owner for FR-001, SEC-001.

Status: proposed

Evidence ids: E-002; E-004

Requirement ids: FR-001; SEC-001

Component name: Access Control Engine

Responsibility: Students and guests may access the internet but must not access the internal staff network. Security operators may review and acknowledge alerts but may not change access policies.

Manual Isolation Module

Detail: Proposed logical owner for FR-002, FR-003.

Status: proposed

Evidence ids: E-003; E-007

Requirement ids: FR-002; FR-003

Component name: Manual Isolation Module

Responsibility: Network administrators may manually isolate a device and restore its access. The system shall not automatically change firewall rules or isolate devices.

Audit Logging Service

Detail: Proposed logical owner for SEC-002.

Status: proposed

Evidence ids: E-005

Requirement ids: SEC-002

Component name: Audit Logging Service

Responsibility: The system shall record administrator policy changes and retain audit records for 30 days.

Open issues: The planning population is 2000 registered devices; concurrency is not yet decided.; Budget, delivery schedule, response-time thresholds and mandatory regulatory frameworks are undecided and must remain open decisions.

Data flows

Proposed source-defined interaction

Detail: Proposed directed path for FR-001.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-001

From component: Actor: Students and guests

To component: Access Control Engine

Data or action: Students and guests may access the internet but must not access the internal staff network.

Proposed source-defined interaction

Detail: Proposed directed path for FR-002.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-002

From component: Actor: Network administrators

To component: Manual Isolation Module

Data or action: Network administrators may manually isolate a device and restore its access.

Proposed source-defined interaction

Detail: Proposed directed path for SEC-001.

Status: proposed

Evidence ids: E-004

Requirement ids: SEC-001

From component: Actor: Security operators

To component: Access Control Engine

Data or action: Security operators may review and acknowledge alerts but may not change access policies.

Proposed source-defined interaction

Detail: Proposed directed path for SEC-002.

Status: proposed

Evidence ids: E-005

Requirement ids: SEC-002

From component: Actor: Administrator

To component: Audit Logging Service

Data or action: The system shall record administrator policy changes and retain audit records for 30 days.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-001

From component: Access Control Engine

To component: Resource: internet

Data or action: Permitted access decision. Students and guests may access the internet but must not access the internal staff network.

Proposed source-defined interaction

Detail: Proposed directed path for FR-001.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-001

From component: Access Control Engine

To component: Resource: internal staff network

Data or action: Denied access attempt. Students and guests may access the internet but must not access the internal staff network.

Trust boundaries

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-001.

Status: proposed

Evidence ids: E-002

Requirement ids: FR-001

Untrusted component: Actor: Students and guests

Trusted component: Access Control Engine

Enforcement: Students and guests may access the internet but must not access the internal staff network.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for FR-002.

Status: proposed

Evidence ids: E-003

Requirement ids: FR-002

Untrusted component: Actor: Network administrators

Trusted component: Manual Isolation Module

Enforcement: Network administrators may manually isolate a device and restore its access.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for SEC-001.

Status: proposed

Evidence ids: E-004

Requirement ids: SEC-001

Untrusted component: Actor: Security operators

Trusted component: Access Control Engine

Enforcement: Security operators may review and acknowledge alerts but may not change access policies.

Proposed source-defined trust boundary

Detail: Proposed logical trust-domain crossing for SEC-002.

Status: proposed

Evidence ids: E-005

Requirement ids: SEC-002

Untrusted component: Actor: Administrator

Trusted component: Audit Logging Service

Enforcement: The system shall record administrator policy changes and retain audit records for 30 days.

9. Threats

Threats

Id: THR-001

Asset: Source-defined permissions and restricted resources

Actor: Unconfirmed actor or cause (planning hypothesis)

Scenario: A principal attempts to bypass a source-defined permission boundary. The boundary under analysis is: Students and guests may access the internet but must not access the internal staff network.

Impact: Unauthorized access or modification outside the stated permission boundary could occur.

Control requirement ids: FR-001

Evidence ids: E-002

Residual risk: Linked obligations define intended behavior, not proven prevention. Enforcement, misuse of permitted access and operational assumptions remain unverified until design and acceptance evidence are reviewed.

Id: THR-002

Asset: Source-defined permissions and restricted resources

Actor: Unconfirmed actor or cause (planning hypothesis)

Scenario: A principal attempts to bypass a source-defined permission boundary. The boundary under analysis is: Security operators may review and acknowledge alerts but may not change access policies.

Impact: Unauthorized access or modification outside the stated permission boundary could occur.

Control requirement ids: SEC-001

Evidence ids: E-004

Residual risk: Linked obligations define intended behavior, not proven prevention. Enforcement, misuse of permitted access and operational assumptions remain unverified until design and acceptance evidence are reviewed.

Id: THR-003

Asset: Source-protected network state

Actor: Unconfirmed actor or cause (planning hypothesis)

Scenario: An implementation defect or misuse triggers an excluded automatic action, contrary to the explicit source boundary: Do not automatically change firewall rules or isolate devices.

Impact: Network state could change contrary to the stakeholder's explicit exclusion.

Control requirement ids: FR-003

Evidence ids: E-007

Residual risk: Linked obligations define intended behavior, not proven prevention. Enforcement, misuse of permitted access and operational assumptions remain unverified until design and acceptance evidence are reviewed.

Assumptions

1. These are defensive planning hypotheses, not observed incidents or approved extra controls. Unrecognized threats and actual implementation effectiveness require human review.

10. Risks

Risks

Id: RISK-001

Description: Implementation may fail to enforce the boundary concerning Source-defined permissions and restricted resources.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: A principal attempts to bypass a source-defined permission boundary. The boundary under analysis is: Students and guests may access the internet but must not access the internal staff network.

Mitigation: Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-001

Evidence ids: E-002

Owner: TBD

Id: RISK-002

Description: Implementation may fail to enforce the boundary concerning Source-defined permissions and restricted resources.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: A principal attempts to bypass a source-defined permission boundary. The boundary under analysis is: Security operators may review and acknowledge alerts but may not change access policies.

Mitigation: Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: SEC-001

Evidence ids: E-004

Owner: TBD

Id: RISK-003

Description: Implementation may fail to enforce the boundary concerning Source-protected network state.

Likelihood: unknown

Impact: unknown

Rationale: This is a planning risk linked to the defensive hypothesis: An implementation defect or misuse triggers an excluded automatic action, contrary to the explicit source boundary: Do not automatically change firewall rules or isolate devices.

Mitigation: Verify the linked canonical obligations against their authored acceptance criteria in an authorized isolated environment. Record gaps and obtain a scope decision before adding any product behavior; preserve explicit exclusions.

Requirement ids: FR-003

Evidence ids: E-007

Owner: TBD

11. User stories and use cases

User stories and use cases

Id: UC-001

Actor: Students and guests

Goal: Students and guests may access the internet but must not access the internal staff network.

Story: As Students and guests, I need the following behavior: Students and guests may access the internet but must not access the internal staff network.

Preconditions

1. A student or guest is authenticated and connected to the wireless network using a valid public identity.

Steps

1. The user attempts to access the internet via standard web browsing services.

Success outcome: The system allows successful access to publicly available internet content.

Requirement ids: FR-001

Evidence ids: E-002

Unresolved decisions: Whether the system must enforce this boundary at the application layer or only via network-level filtering.; The specific mechanism for detecting and blocking attempts to reach internal staff network resources.

Id: UC-002

Actor: Students and guests

Goal: Students and guests may access the internet but must not access the internal staff network.

Story: As Students and guests, I need the following behavior: Students and guests may access the internet but must not access the internal staff network.

Preconditions

1. A student or guest is authenticated and connected to the wireless network using a valid public identity.

Steps

1. The user attempts to access internal staff network resources via IP address or domain name (e.g., internal.staff.company.com).

Success outcome: The system denies access and no product message is returned; traffic is blocked as evidenced by an external test capture of the network response state.

Requirement ids: FR-001

Evidence ids: E-002

Unresolved decisions: Whether the system must enforce this boundary at the application layer or only via network-level filtering.; The specific mechanism for detecting and blocking attempts to reach internal staff network resources.

Id: UC-003

Actor: Network administrators

Goal: Network administrators may manually isolate a device and restore its access.

Story: As Network administrators, I need the following behavior: Network administrators may manually isolate a device and restore its access.

Preconditions

1. A network administrator logs into the system with valid administrative credentials.

Steps

1. The administrator selects a device to isolate via the manual isolation interface.

Success outcome: The system isolates the device from wireless access and updates the access control state accordingly.

Requirement ids: FR-002

Evidence ids: E-003

Unresolved decisions: The specific interface or UI component used for isolation and restoration is not defined; this remains an implementation parameter.

Id: UC-004

Actor: Network administrators

Goal: Network administrators may manually isolate a device and restore its access.

Story: As Network administrators, I need the following behavior: Network administrators may manually isolate a device and restore its access.

Preconditions

1. A network administrator logs into the system with valid administrative credentials.

Steps

1. The administrator selects a device that has been isolated to restore access via the manual restoration interface.

Success outcome: The system restores wireless access for the device and updates the access control state accordingly.

Requirement ids: FR-002

Evidence ids: E-003

Unresolved decisions: The specific interface or UI component used for isolation and restoration is not defined; this remains an implementation parameter.

Id: UC-005

Actor: Security operators

Goal: Security operators may review and acknowledge alerts but may not change access policies.

Story: As Security operators, I need the following behavior: Security operators may review and acknowledge alerts but may not change access policies.

Preconditions

1. A security operator accesses an alert dashboard

Steps

1. The operator views a network anomaly report

Success outcome: The operator can acknowledge the alert and receive confirmation of acknowledgment

Alternate flows

1. Given: A security operator receives an alert with a policy violation. When: The operator attempts to modify an access policy through the system interface. Then: The system denies the modification request and returns a response indicating that policy modifications are not permitted for operators. Evidence: Capture of the system's direct response shows a denial of the modification action, with no change to the access policies in storage or configuration.

Requirement ids: SEC-001

Evidence ids: E-004

Id: UC-006

Actor: Security operators

Goal: Security operators may review and acknowledge alerts but may not change access policies.

Story: As Security operators, I need the following behavior: Security operators may review and acknowledge alerts but may not change access policies.

Preconditions

1. An operator receives an alert with a visible status indicator.

Steps

1. The operator reviews the alert and acknowledges it via the system interface.

Success outcome: The system updates the alert status to 'acknowledged' and retains the original policy unchanged; no policy modification occurs.

Requirement ids: SEC-001

Evidence ids: E-004

Open issues: Whether the system must enforce this boundary at the application layer or only via network-level filtering.; The specific mechanism for detecting and blocking attempts to reach internal staff network resources.; The specific interface or UI component used for isolation and restoration is not defined; this remains an implementation parameter.

12. Verification and acceptance

Tests

Id: TEST-001

Requirement ids: FR-001

Procedure

1. Given: A student or guest is authenticated and connected to the wireless network using a valid public identity.
2. When: The user attempts to access the internet via standard web browsing services.

Expected result: Expected: The system allows successful access to publicly available internet content.

Evidence to record: Evidence: Record of successful page load from a public website, such as google.com.

Test type: positive

Id: TEST-002

Requirement ids: FR-001

Procedure

1. Given: A student or guest is authenticated and connected to the wireless network using a valid public identity.
2. When: The user attempts to access internal staff network resources via IP address or domain name (e.g., internal.staff.company.com).

Expected result: Expected: The system denies access and no product message is returned; traffic is blocked as evidenced by an external test capture of the network response state.

Evidence to record: Evidence: Export of a test-instrumented network response state capture showing denied access to internal staff network resources, captured at the time of request and verified against known firewall rule enforcement.

Test type: inspection

Id: TEST-003

Requirement ids: FR-002

Procedure

1. Given: A network administrator logs into the system with valid administrative credentials.
2. When: The administrator selects a device to isolate via the manual isolation interface.

Expected result: Expected: The system isolates the device from wireless access and updates the access control state accordingly.

Evidence to record: Evidence: Capture the system response state after manual isolation via API call and validate it against expected access control update.

Test type: positive

Id: TEST-004

Requirement ids: FR-002

Procedure

1. Given: A network administrator logs into the system with valid administrative credentials.
2. When: The administrator selects a device that has been isolated to restore access via the manual restoration interface.

Expected result: Expected: The system restores wireless access for the device and updates the access control state accordingly.

Evidence to record: Evidence: Capture the system response state after manual restoration of wireless access for the device, including the updated access control state and the timestamp of the action.

Test type: positive

Id: TEST-005

Requirement ids: FR-003

Procedure

1. Given: The system is configured to automatically modify firewall rules or isolate a device in response to an event
2. When: An automated action such as rule modification or isolation is initiated by the system

Expected result: Expected: The action shall be denied and no change to firewall configuration or device state shall occur

Evidence to record: Evidence: Capture of unchanged firewall configuration and unmodified device state before and after the attempted action

Test type: negative

Id: TEST-006

Requirement ids: SEC-001

Procedure

1. Given: A security operator accesses an alert dashboard
2. When: The operator views a network anomaly report

Expected result: Expected: The operator can acknowledge the alert and receive confirmation of acknowledgment

Evidence to record: Evidence: Record of acknowledged alert with timestamp and operator identity

Test type: inspection

Id: TEST-007

Requirement ids: SEC-001

Procedure

1. Given: An operator receives an alert with a visible status indicator.
2. When: The operator reviews the alert and acknowledges it via the system interface.

Expected result: Expected: The system updates the alert status to 'acknowledged' and retains the original policy unchanged; no policy modification occurs.

Evidence to record: Evidence: Capture of the system response showing alert status updated to ' acknowledged' with no change in access policies, as verified by a test-instrumented state capture of the alert object's properties.

Test type: positive

Id: TEST-008

Requirement ids: SEC-001

Procedure

1. Given: A security operator receives an alert with a policy violation.
2. When: The operator attempts to modify an access policy through the system interface.

Expected result: Expected: The system denies the modification request and returns a response indicating that policy modifications are not permitted for operators.

Evidence to record: Evidence: Capture of the system's direct response shows a denial of the modification action, with no change to the access policies in storage or configuration.

Test type: negative

Id: TEST-009

Requirement ids: SEC-002

Procedure

1. Given: An administrator modifies a network access policy, such as changing authentication requirements or access permissions.
2. When: The system records the change and stores it in an audit log.

Expected result: Expected: The audit record is retained for 30 days and remains accessible for review by authorized personnel.

Evidence to record: Evidence: Inspect the audit log to confirm that a policy change event was recorded with timestamp, actor identity, action type, and before/after state.

Test type: inspection

Limitations: These verification plans have not been executed; their acceptance outcomes require testing.

13. Glossary

Summary: Source-present terminology; definitions introduce no additional product obligation.

Items

Access control

Detail: Rules and enforcement governing which principals may perform actions on resources.

Status: proposed

Evidence ids: E-001

Policy

Detail: A rule or decision governing permitted actions; its project-specific meaning remains defined by the source requirements.

Status: proposed

Evidence ids: E-004; E-005

Audit record

Detail: A stored record of an event. Required event scope and retention are determined by the linked source, not this definition.

Status: proposed

Evidence ids: E-005

Device isolation

Detail: Restriction of a device's network access; the source determines whether and by whom this action is permitted.

Status: proposed

Evidence ids: E-003; E-007

Access restoration

Detail: Returning access that was previously restricted, within the source-defined permission boundary.

Status: proposed

Evidence ids: E-003

Concurrency

Detail: Activity occurring at the same time; it is distinct from the registered planning population.

Status: proposed

Evidence ids: E-006

Registered population

Detail: The inventory of registered users or devices; it does not by itself establish a simultaneous load or a maximum.

Status: proposed

Evidence ids: E-006

Retention

Detail: The period or rule for keeping records. A bare duration does not itself specify deletion timing or minimum/maximum bounds.

Status: proposed

Evidence ids: E-005

Firewall

Detail: A control that permits or restricts network traffic according to rules; mentioning it does not authorize automatic changes.

Status: proposed

Evidence ids: E-007

14. SDLC recommendation and workflow

Selected model: Agile

Justification: The evolving stability (E-008) and frequent feedback (E-008) favor Agile because it enables iterative delivery, continuous review, and rapid adaptation to changing requirements. The score of 90 indicates strong suitability for this context.

Tradeoffs

1. Recurring delivery requires stakeholder review and backlog refinement; a cycle does not guarantee completeness or remove unresolved decisions.

Phases

Name: Requirements and verification planning

Phase type: planning

Activities

1. Review and baseline the source-defined work group: FR-001: Students and guests may access the internet but must not access the internal staff network.; FR-002: Network administrators may manually isolate a device and restore its access.; FR-003: The system shall not automatically change firewall rules or isolate devices.
2. Review and baseline the source-defined work group: SEC-001: Security operators may review and acknowledge alerts but may not change access policies.; SEC-002: The system shall record administrator policy changes and retain audit records for 30 days.
3. Create the unit, integration, system and acceptance test plan before implementation; retain each authored acceptance criterion.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Versioned requirements baseline
2. Unit, integration, system and acceptance test plan
3. Unresolved-decision record

Entry criteria

1. Authoritative inputs and canonical requirements are available for review.

Exit criteria

1. A human reviewer has recorded the baseline decision and disposition of unresolved scope.

Security checks

1. Review implementation and acceptance evidence against FR-001: Students and guests may access the internet but must not access the internal staff network.
2. Review implementation and acceptance evidence against FR-002: Network administrators may manually isolate a device and restore its access.
3. Review implementation and acceptance evidence against FR-003: The system shall not automatically change firewall rules or isolate devices.
4. Review implementation and acceptance evidence against SEC-001: Security operators may review and acknowledge alerts but may not change access policies.

5. Review implementation and acceptance evidence against SEC-002: The system shall record administrator policy changes and retain audit records for 30 days.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; SEC-001; SEC-002

Name: Agile delivery cycle

Phase type: iteration

Activities

1. Review and prioritize the backlog for the next iteration without changing source-defined scope.
2. Implement the source-defined work group: FR-001: Students and guests may access the internet but must not access the internal staff network.; FR-002: Network administrators may manually isolate a device and restore its access.; FR-003: The system shall not automatically change firewall rules or isolate devices.
3. Implement the source-defined work group: SEC-001: Security operators may review and acknowledge alerts but may not change access policies.; SEC-002: The system shall record administrator policy changes and retain audit records for 30 days.
4. In an authorized isolated test environment, execute TEST-001: Given: A student or guest is authenticated and connected to the wireless network using a valid public identity. When: The user attempts to access the internet via standard web browsing services. Expected: The system allows successful access to publicly available internet content.
5. In an authorized isolated test environment, execute TEST-002: Given: A student or guest is authenticated and connected to the wireless network using a valid public identity. When: The user attempts to access internal staff network resources via IP address or domain name (e.g., internal.staff.company.com). Expected: The system denies access and no product message is returned; traffic is blocked as evidenced by an external test capture of the network response state.
6. In an authorized isolated test environment, execute TEST-003: Given: A network administrator logs into the system with valid administrative credentials. When: The administrator selects a device to isolate via the manual isolation interface. Expected: The system isolates the device from wireless access and updates the access control state accordingly.
7. In an authorized isolated test environment, execute TEST-004: Given: A network administrator logs into the system with valid administrative credentials. When: The administrator selects a device that has been isolated to restore access via the manual restoration interface. Expected: The system restores wireless access for the device and updates the access control state accordingly.
8. In an authorized isolated test environment, execute TEST-005: Given: The system is configured to automatically modify firewall rules or isolate a device in response to an event When: An automated action such as rule modification or isolation is initiated by the system Expected: The action shall be denied and no change to firewall configuration or device state shall occur
9. In an authorized isolated test environment, execute TEST-006: Given: A security operator accesses an alert dashboard When: The operator views a network anomaly report Expected: The operator can acknowledge the alert and receive confirmation of acknowledgment
10. In an authorized isolated test environment, execute TEST-007: Given: An operator receives an alert with a visible status indicator. When: The operator reviews the alert and acknowledges it via the system interface. Expected: The system updates the alert status to 'acknowledged' and retains the original policy unchanged; no policy modification occurs.
11. In an authorized isolated test environment, execute TEST-008: Given: A security operator receives an alert with a policy violation. When: The operator attempts to modify an access policy

through the system interface. Expected: The system denies the modification request and returns aresponse indicating that policy modifications are not permitted for operators.

12. In an authorized isolated test environment, execute TEST-009: Given: An administrator modifies a network access policy, such as changing authentication requirements or access permissions. When: The system records the change and stores it in an audit log. Expected: The audit record is retained for 30 days and remains accessible for review by authorized personnel.
13. Collect stakeholder feedback in accordance with the source: Requirements evolve frequently and stakeholders provide weekly stakeholder feedback.
14. Return accepted stakeholder feedback and observed defects to the backlog and plan the next iteration.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirement-linked implementation increment
2. Recorded test evidence
3. Reviewed feedback and next-iteration backlog

Entry criteria

1. Prior work, source scope and the reviewed test plan are available for iteration planning.

Exit criteria

1. A human reviewer has assessed the increment, test evidence and feedback; unresolved changes return to the next iteration.

Security checks

1. Review implementation and acceptance evidence against FR-001: Students and guests may access the internet but must not access the internal staff network.
2. Review implementation and acceptance evidence against FR-002: Network administrators may manually isolate a device and restore its access.
3. Review implementation and acceptance evidence against FR-003: The system shall not automatically change firewall rules or isolate devices.
4. Review implementation and acceptance evidence against SEC-001: Security operators may review and acknowledge alerts but may not change access policies.
5. Review implementation and acceptance evidence against SEC-002: The system shall record administrator policy changes and retain audit records for 30 days.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; SEC-001; SEC-002

Name: Release and handoff

Phase type: release

Activities

1. Review the exact requirements baseline, implementation links and verification evidence before a release decision.
2. Carry unresolved source decisions into handoff; generated text is not human approval.

Roles: Proposed project owner; Proposed development lead; Proposed verification reviewer

Deliverables

1. Requirements and verification evidence package
2. Pending human release decision
3. Outstanding-decision record

Entry criteria

1. Verification evidence and unresolved decisions are available for human review.

Exit criteria

1. A human has recorded the release/handoff decision for the reviewed content.

Security checks

1. Review implementation and acceptance evidence against FR-001: Students and guests may access the internet but must not access the internal staff network.
2. Review implementation and acceptance evidence against FR-002: Network administrators may manually isolate a device and restore its access.
3. Review implementation and acceptance evidence against FR-003: The system shall not automatically change firewall rules or isolate devices.
4. Review implementation and acceptance evidence against SEC-001: Security operators may review and acknowledge alerts but may not change access policies.
5. Review implementation and acceptance evidence against SEC-002: The system shall record administrator policy changes and retain audit records for 30 days.

Human approval: Pending human decision by Proposed project owner on this phase's deliverables and unresolved decisions; no approval is recorded by generation.

Requirement ids: FR-001; FR-002; FR-003; SEC-001; SEC-002

Unknown factors: budget; delivery; failure_consequences; legacy; regulation; schedule; uncertainty; verification

SDLC rankings and decision evidence

Policy: sequential-sdlc-1.1

Generation profile: sequential-sdlc-generation-2.0

Factors

Name: stability

Value: evolving

Evidence ids: E-008

Name: feedback

Value: frequent

Evidence ids: E-008

Ranked candidates

Model: Agile

Score: 90

Reasons: stability=evolving (E-008); feedback=frequent (E-008)

Model: Hybrid

Score: 72

Reasons: stability=evolving (E-008); feedback=frequent (E-008)

Model: Incremental

Score: 60

Reasons: stability=evolving (E-008)

Model: DevSecOps

Score: 40

Reasons: No differentiating input evidence

Model: Spiral

Score: 40

Reasons: No differentiating input evidence

Model: V-Model

Score: 40

Reasons: No differentiating input evidence

Model: Waterfall

Score: 15

Reasons: stability=evolving (E-008)

Evidence register

Id: E-001

Source: description

Quote: Build a campus wireless network access-control requirements specification.

Id: E-002

Source: description

Quote: Students and guests may access the internet but must not access the internal staff network.

Id: E-003

Source: description

Quote: Network administrators may manually isolate a device and restore its access.

Id: E-004

Source: description

Quote: Security operators may review and acknowledge alerts but may not change access policies.

Id: E-005

Source: description

Quote: The system must record administrator policy changes and retain audit records for 30 days.

Id: E-006

Source: description

Quote: The planning population is 2000 registered devices; concurrency is not yet decided.

Id: E-007

Source: description

Quote: Do not automatically change firewall rules or isolate devices.

Id: E-008

Source: description

Quote: Requirements evolve frequently and stakeholders provide weekly stakeholder feedback.

Id: E-009

Source: description

Quote: Budget, delivery schedule, response-time thresholds and mandatory regulatory frameworks are undecided and must remain open decisions.

Source references

kb-c1e0239af9dd46bd:1 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 1

Page or section: text chunk 2

Relevance score: 0.7246

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:8 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 8

Page or section: text chunk 9

Relevance score: 0.6986

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-788cd8fba7ce0acb:4 - 03_OWASP_CheatSheets/083_OWASP_Network_Segmentation_Cheat_Sheet.md

Source document id: kb-788cd8fba7ce0acb

Chunk index: 4

Page or section: text chunk 5

Relevance score: 0.7011

Sha256: 2b91c4e103da8e53c98e3b6006b23c2bb02e8c55e5805a56f0068c7b31a477d8

kb-c1e0239af9dd46bd:11 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 11

Page or section: text chunk 12

Relevance score: 0.6915

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-c1e0239af9dd46bd:17 - 03_OWASP_CheatSheets/107_OWASP_Zero_Trust_Architecture_Cheat_Sheet.md

Source document id: kb-c1e0239af9dd46bd

Chunk index: 17

Page or section: text chunk 18

Relevance score: 0.6875

Sha256: f5fe6d1e1cfefc05da1b1b9773acee6b7e9d156f4aa69ab0a14a7a1adfd2a8ef

kb-0150d5352d0ec304:22 - 03_OWASP_CheatSheets/036_OWASP_Authentication_Cheat_Sheet.md

Source document id: kb-0150d5352d0ec304

Chunk index: 22

Page or section: text chunk 23

Relevance score: 0.6928

Sha256: b1d318a57f742d12b57bccda3153873653652c767038c5ca3d9d74350819f303

Validation findings

Code: human_semantic_review_required

Severity: warning

Section: document

Message: Review completeness, actor permissions, contradictions and citation applicability; schema and link checks do not prove semantic correctness.

Human review history

None recorded.

AI drafting record

Requested model: cybersrs-qwen3-4b-ft:latest

Actual models used: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Section models

Introduction: cybersrs-qwen3-4b-ft:latest

Functional requirements: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Non-functional requirements: cybersrs-qwen3-4b-ft:latest

Security, privacy and compliance: cybersrs-qwen3-4b-ft:latest; qwen3:4b-instruct-2507-q4_K_M

Data requirements: cybersrs-qwen3-4b-ft:latest

Network and interfaces: cybersrs-qwen3-4b-ft:latest

Compliance and policy applicability matrix: cybersrs-qwen3-4b-ft:latest

Architecture: cybersrs-qwen3-4b-ft:latest

Threats: cybersrs-qwen3-4b-ft:latest

Risks: cybersrs-qwen3-4b-ft:latest

User stories and use cases: cybersrs-qwen3-4b-ft:latest

Verification and acceptance: cybersrs-qwen3-4b-ft:latest

Glossary: cybersrs-qwen3-4b-ft:latest

SDLC recommendation and workflow: cybersrs-qwen3-4b-ft:latest

Prompt version: sequential-1.27

Requirements traceability matrix

Requirement	Evidence / sources	Stories / tests / SDLC
FR-001	E-002	UC-001, UC-002, TEST-001, TEST-002, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-002	E-003	UC-003, UC-004, TEST-003, TEST-004, Requirements and verification planning, Agile delivery cycle, Release and handoff
FR-003	E-007	TEST-005, Requirements and verification planning, Agile delivery cycle, Release and handoff
SEC-001	E-004	UC-005, UC-006, TEST-006, TEST-007, TEST-008, Requirements and verification planning, Agile delivery cycle, Release and handoff
SEC-002	E-005	TEST-009, Requirements and verification planning, Agile delivery cycle, Release and handoff

Model: cybersrs-qwen3-4b-ft:latest + qwen3:4b-instruct-2507-q4_K_M; prompt: sequential-1.27